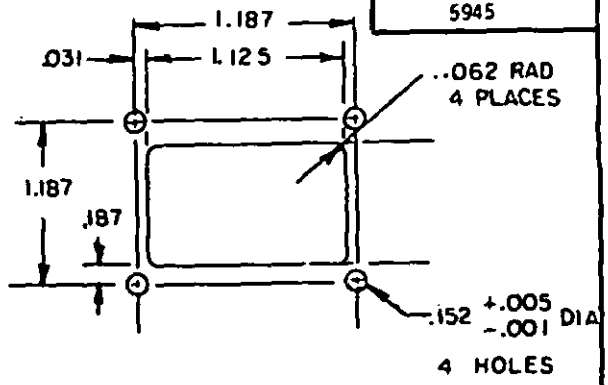
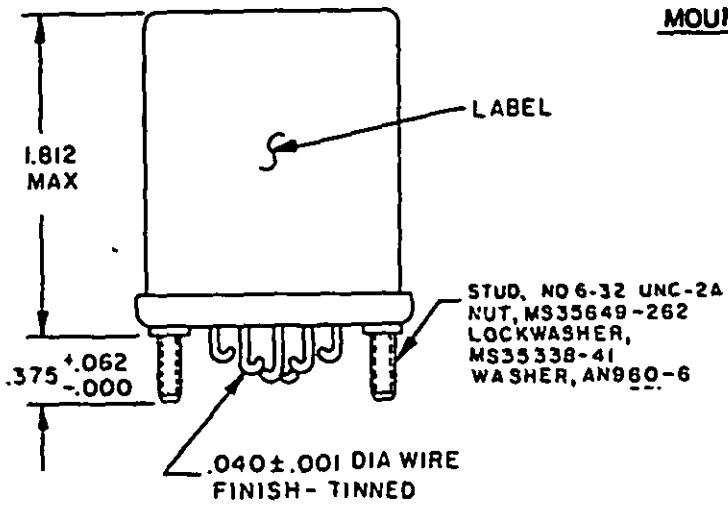
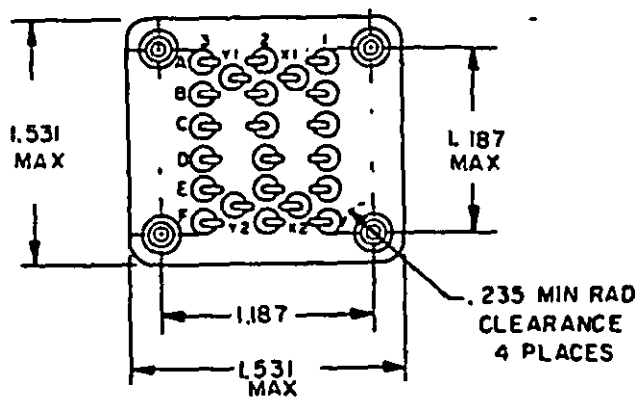
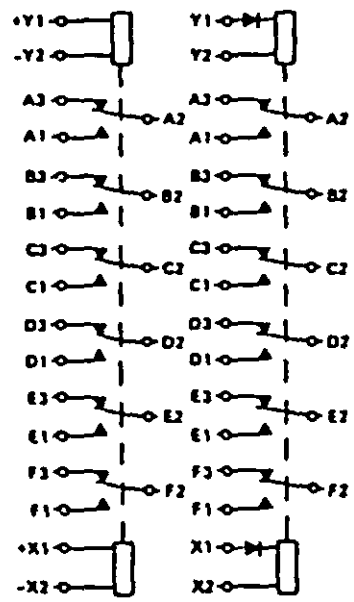
FED. SUP CLASS
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MOUNTING HOLE LAYOUT




CIRCUIT - D1
(SEE NOTES
4 AND 6

CIRCUIT - A1
(SEE NOTES
5, 6, AND 9)

APPROVED 1 March 1963 REVISED Ⓓ 30 SEPT 1987

Ⓓ ENTIRE STANDARD REVISED

| | | | |
|---|---|---|---|
| P.A AF - 85<br>Other Cust Navy - AS | International Interest | TITLE RELAYS, ELECTROMAGNETIC, 6PDT, 5 AMPERES, TYPE I, MAGNETIC LATCH, STUD MOUNTED, SOLDER HOOK TERMINALS, HERMETICALLY SEALED | MILITARY STANDARD<br>MS25469 |
| Procurement Specification MIL-R-6106 | | SUPERSEDES: | PAGE 1 OF 5 |

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. 5945-0767-04

AMSC N/A

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

User activities: Army - Navy - Air Force -

Review activities: Army - Navy - EC Air Force - 11, 99

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| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .375 | 9.53 |
| .005 | 0.13 | .812 | 20.62 |
| .031 | 0.79 | 1.125 | 28.58 |
| .040 | 1.02 | 1.187 | 30.15 |
| .062 | 1.57 | 1.531 | 38.89 |
| .138 | 3.51 | 1.812 | 46.02 |
| .152 | 3.86 | | |
| .235 | 5.95 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Caution note to observe polarity must appear on relays with dc coils.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. Relay is magnetically latched in both positions.
7. Shock, vibration, and acceleration requirement applicable with coils de-energized.
8. Terminal numbers shall not appear on relay header. There shall be affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified herein.
8. Pins to be perpendicular to header surface within one degree.
9. MS25469-A1 is inactive for new design.
10. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
11. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics. 1/

| Dash number MS25469- | Type | Coil | Terminal type | Mounting or mating socket | Auxiliary contacts | Max weight in pounds |
|---|---|---|---|---|---|---|
| 01 | I | dc | Solder hook | Stud | None | 0.37 |
| A1 2/ | I | ac | Solder hook | Stud | None | 0.38 |

1/ MS25469-A01 cancelled without replacement.
2/ MS25469-A1 inactive for new design.

APPROVED 1 March 1963 REVISED (1) ENTIRE STANDARD REVISED

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11, 99

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Review activities: Army -
Navy - EC
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

DD FORM 1 MAY 73 672 (Coordinated) supersedes previous editions of this form are obsolete.

Procurement Specification MIL-R-6106

P.A
AF - 85
Other Cust
Navy - AS

Interchangeability

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 6PDT, 5 AMPERES, TYPE 1, MAGNETIC LATCH, STUD MOUNTED, SOLDER HOOK TERMINALS, HERMETICALLY SEALED

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MS25469

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TABLE II. Operating characteristics.

| MS part no. MS25469- | Coil data | | | | | | | | | | Time - (milliseconds maximum) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Operate 3/ | Release 4/ | Contact bounce | | | |
| | | Volts 1/ | Freq. Hz | Res Ω | Volts | Amperes | Normal 2/ | High temp test | Cont current test | | | Main | | Aux | |
| | | | | | | | | | | | | NO | NC | NO | NC |
| D1 | X1, X2 Y1, Y2 | 28 | dc | | 29 | 0.17 | 18 | 18 | 19.8 | 25 | N/A | 2 | 2 | | |
| A1 6/ | X1, X2 Y1, Y2 | 115 | 400 5/ | | 122 | 0.07 | 90 | 90 | 95 | 25 | N/A | 2 | 2 | | |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25469-A1 may be used on 60 Hz if maximum ambient temperature is limited to +85°C, maximum current coil will be 0.077 ampere.
6/ MS25469 - A1 is inactive for new design.

APPROVED 1 March 1963 REVISED (D) ENTIRE STANDARD REVISED

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Review activities: Army -
Navy - EC
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

TABLE III. Rated contact load (amperes per pole) (case grounded).

| Type of load | Life operating cycles X $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 100 | 5 | 5 | | | 5 | 4 | | | | | | | |
| Inductive | 100 | | | | | | | | | | | | | |
| Inductive | 20 | 3 | 3 | | | 3 | 2 | | | | | | | |
| Motor | 100 | 1.5 | 1.5 | | | 1.5 | 1 | | | | | | | |
| Lamp | 100 | 0.8 | 2 | | | 0.8 | 0.6 | | | | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 400 | 1.25 | 1.25 | | | 1.25 | 1 | | | | | | | |
| Intmd current | | Applicable per specification | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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P.A
AF - 85
Other Cust
Navy - AS

Procurement Specification
MIL-R-6106

International interest

SUPERSEDES:

TITLE
RELAYS, ELECTROMAGNETIC, 6PDT
5 AMPERES, TYPE 1, MAGNETIC LATCH,
STUD MOUNTED, SOLDER HOOK TERMINALS,
HERMETICALLY SEALED.

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Review activities: Army -
Navy - EC
Air Force - 11, 99

User activities: Army -
Navy -
Air Force -

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P.A
AF - 85
Other Cust
Navy - AS

Procurement Specification
MIL-R-6106

Superseding Information

SUPERSEDES:

TITLE
RELAYS, ELECTROMAGNETIC, 6PDT
5 AMPERES, TYPE I, MAGNETIC LATCH,
STUD MOUNTED, SOLDER HOOK TERMINALS,
HERMETICALLY SEALED.

MILITARY STANDARD

MS25469

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FED. SUP CLASS
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## Environmental characteristics

| | |
|---|---|
| Temperature range | -70°C to +125°C |
| Max altitude rating | 80,000 ft |
| Shock G-level | 50 g's |
| Duration | 11 ms |
| Max duration contact opening | 10 µs |
| Vibration - Sinusoidal | |
| G-level | 10 g's |
| Frequency range | 5 - 1,500 Hz |
| Acceleration | 15 g's |

## Quality conformance inspection

Performance of groups B and C tests are not applicable to MS25469-A1.

## Electrical characteristics

| | |
|---|---|
| Insulation resistance, initial | 100 megohms |
| After life or environmental tests | 50 megohms |

Dielectric strength (sea level)

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 V rms | 1,000 V rms |
| Aux contacts | | |
| All other points | 1,000 V rms | 1,000 V rms |

Dielectric strength (altitude)

| | 80,000 ft |
|---|---|
| Coil to case | 250 V rms |
| Aux contacts | |
| All other points | 250 V rms |

| | |
|---|---|
| Max contact drop initial | 0.150 volt |
| After life test | 0.175 volt |
| Overload current | 20 amperes |
| Rupture current | 25 amperes |
| Duty rating | Continuous |
| RFI specification | MIL-STD-461 |

(Applicable to coil circuits of ac operated relays)

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